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Project Title:	Dynamics of auxin transport protein localization and gravitropism		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
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Human Research Program Elements:	None		
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Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
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Space Biology Special Category:	None		
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Comments:			
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No. of PhD Candidates:		**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Plants respond to gravity in numerous ways; the best understood is the change in growth orientation in response to reorientation of plants relative to the gravity vector. A growth asymmetry develops, which allows the plant organ to return to the original angle of growth relative to gravity. Understanding gravitropism is essential to facilitate growth of plants in the microgravity environment of space. Gravitropic bending is mediated, at least in part, by redistribution of the plant hormone auxin. This proposal examines the mechanisms by which the cellular asymmetry of auxin transport proteins is established and the molecular mechanisms that allow translocation of the auxin signals from the site of perception to the site of reaction.
We will employ three approaches to transiently alter the expression and localization of auxin transport proteins to examine the effect on gravitropic curvature in roots of the model species Arabidopsis thaliana. First, we will use a

Task Description:	temperature sensitive mutant, scd1-1, with a defect in endomembrane targeting, to transiently disrupt, auxin transport protein localization and gravitropic curvature. Second, we will transiently perturb the activity and/or localization of auxin transport proteins through a pharmacological approach and measure the resulting effect on gravitropic kinetics. In our third approach, we will use well characterized null mutants with defects in auxin transport proteins and will express wild type copies of defective genes in an inducible expression system. This approach of reintroducing a gene function at a controlled time points during the gravitropic process, should allow us to understand when these proteins function in controlling auxin transport and gravitropism. The planned experiments will combine high resolution laser scanning confocal microscopy and high temporal resolution gravitropic analyses with transient perturbation of auxin transport to understand the dynamics of auxin transport protein localization and resulting auxin asymmetries that drive differential gravitropic growth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Our initial focus has been on examination of the scd1-1 mutant and we have asked if IAA transport, the distribution of IAA efflux proteins, and gravity response are altered in the scd1-1 mutant, which has a defect in a gene encoding a protein predicted to function in Rab GTPase-dependent protein trafficking. We monitored gravitropic curvature, growth rate, and auxin transport in scd1-1 as seedlings were grown at the permissive temperature (18°C) or the non-permissive temperature (25°C). At the permissive temperature gravitropism, growth, and auxin transport are at equivalent levels in wild-type and scd1-1. The gravitropic response of scd1-1 seedlings was quantified by high resolution imaging, taking images every 2 minutes for 8 hours to measure the full gravitropic response. At the nonpermissive temperature, this process is profoundly reduced. Second, we examined the transport of auxin in the scd1-1 mutant, to determine if the mechanism by which gravity response is impaired is tied to the ability to transport IAA. Auxin moves in two directions in the root in two distinct tissues. Basipetal or shootward IAA transport, regulates differential gravitropic growth, and acropetal or rootward transport, provides auxin for lateral root development. We examined both polarities of IAA transport and find that both basipetal and acropetal transport are reduced in the scd1-1 mutant relative to wild-type at 25°C, while similar transport amounts are detected at 18°C. We also defined the time required after transition from permissive to nonpermissive temperature to lead to the altered auxin transport phenotype. Just 6 hours of incubation at RT (25°C) after growth at 18°C is sufficient to show a difference in acropetal IAA transport between scd1-1 and wild-type. We are currently measuring IAA transport in seedlings transitioned between temperatures and correlating the time required to alter gravitropic response to the time required to altered gravity response and lateral root formation, to show a physiological relationship between these two parameters. The third line of experiments in scd1-1 focuses on examination of the effects of temperature changes on the distribution of auxin transport proteins. We have used laser scanning confocal microscopy to examine the tissue level and sub-cellular distribution of auxin efflux carriers PIN1, PIN2, and PIN3 that are translational fusions to GFP. When scd1-1 was grown at the nonpermissive temperature, the expression of these reporters was restricted to fewer cells and was reduced in intensity. The subcellular distribution of these three reporters was similar in wild-type and scd1-1 at 18°C with the expected asymmetrical pattern at the plasma membrane in wild type and scd1-1, consistent with these proteins defining the polarity of auxin transport. At 25°C, the PIN1 and PIN2:GFP fusion in scd1-1 were found at substantially lower levels in the plasma membrane, and accumulated in endomembranes. Yet, the membrane asymmetries of these proteins were generally maintained, although substantially weaker. The apparent structure and character of endomembranes that accumulated PIN1-GFP and PIN2-GFP were quite different. This result suggests that in the mutant, these two auxin transport proteins may be trafficked by different pathways in the different cell types in which they are found, causing them to be shunted to different structures where they accumulate. In contrast, PIN3-GFP looked qualitatively the same in both wild type and scd1-1 at the subcellular level, but was restricted to fewer cells at 25°C compared to wild type. Ongoing experiments focus on identification of the identity of the endomembranes that accumulate these PIN-GFP fusions by crossing mutant lines with fluorescent protein fusions that have distinct organelle locations. A second line of experimentation is utilizing a pharmacological approach to transiently disrupt activity and/or targeting of auxin transport proteins to understand their role in gravitropic responses. We have examined compounds which inhibit protein phosphorylation and dephosphorylation, specifically staurosporine and cantharidin, respectively, and combined these inhibitors with mutants in specific kinase and phosphatase genes, to provide evidence of reciprocal regulation of auxin transport by protein phosphorylation and regulation of gravitropic responses (Sukumar et al., 2009). We are also examining the effect of a range of chemicals that disrupt auxin transport protein targeting and gravity responses isolated in a chemical genomic screen by Glenn Hicks and Natasha Raikhel at UC Riverside. We are focusing on the effect of a subset of these chemicals on auxin transport and the kinetics of root gravitropism. These experiments will first contribute insight into the function of these compounds to identify their targets of action. Second, we will use these compounds to disrupt the gravitropic response by altering auxin transport and then to use these to perturb gravitropic responses to provide insight into the dynamics of this process. Training This grant is currently supporting one graduate student and half of the salary of a post-doctoral researcher, and two undergraduates. The graduate student working on the project, Jonathan Isley, is performing the phenotypic characterization of gravitropic response and auxin transport in the scd1-1 mutant. The post-doctoral researcher, Dr. Tanya Falbel, is performing the crosses needed to examine reporters and look for mechanistic roles of auxin transport downstream of scd1-1. These two researchers are working collaboratively to image the reporter lines using laser scanning confocal microscopy with the goal of understanding the effect of scd1-1 on the localization of auxin transport protein:GFP fusions and endomembrane marker: fluorescent protein fusions. The pharmacological experiments are being performed in collaboration between Jonathan Isley and a graduate student, Michelle Brown, in the laboratory of Natasha Raikhel at the University of California at Riverside, and with a WFU undergraduate, Ryan Hughes. The gravitropic effects of flavonoids, which are endogenous regulators of auxin transport, is being performed by another WFU undergraduate, Stephanie Zhang.
Bibliography Type:	Description: (Last Updated: 08/25/2020)

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Abstracts for Journals and Proceedings	Isley J, Mattox C, Cooper KL, Falbel T, Muday GK. "Altered gravitropism and auxin transport in the scd1 mutant of Arabidopsis thaliana." 25th Annual Meeting of the American Society for Gravitational and Space Biology, Raleigh, NC, November 5-9, 2009. Gravitational and Space Biology 2009 Nov;23(1):7. , Nov-2009
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